

Mastering Quantum Mechanics — Problem 9.2

Harmonic oscillator with a wall

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Problem 9.2

Harmonic oscillator with a wall. Consider a *half* harmonic oscillator described by the potential

$$V(x) = \begin{cases} \frac{1}{2}m\omega^2 x^2, & x > 0, \\ \infty, & x < 0, \end{cases}$$

(a spring that can be stretched but not compressed). Explain how the eigenstates of this system and their energies are related to those of the full harmonic oscillator.

Solution

Strategy: the wall as a boundary condition

Where $V = \infty$ (i.e. $x < 0$) the wave function must vanish, so any eigenstate $\psi(x)$ of the half oscillator satisfies

$$\psi(x) = 0 \quad \text{for } x \leq 0, \quad \text{in particular} \quad \psi(0) = 0, \quad (1)$$

the last line being the continuity requirement at the wall. For $x > 0$ the potential is exactly that of the full oscillator, so on the half-line ψ obeys the *same* time-independent Schrödinger equation as the full oscillator (Eq. (9.1.11)):

$$-\frac{\hbar^2}{2m} \psi''(x) + \frac{1}{2}m\omega^2 x^2 \psi(x) = E \psi(x), \quad x > 0. \quad (2)$$

Key idea. The half-oscillator eigenstates are precisely the full-oscillator energy eigenstates that already satisfy $\psi(0) = 0$, restricted to $x > 0$. The infinite wall does not change the equation on $x > 0$; it only *selects* the solutions with a node at the origin.

Which full-oscillator states qualify? Parity selects the odd ones

The full-oscillator eigenstates are (Eqs. (9.2.35)–(9.2.36))

$$\varphi_n(x) = N_n H_n(x/L_0) e^{-x^2/2L_0^2}, \quad E_n = \hbar\omega\left(n + \frac{1}{2}\right), \quad (3)$$

with $L_0 = \sqrt{\hbar/m\omega}$. The Hermite polynomials have definite parity, $H_n(-u) = (-1)^n H_n(u)$ (e.g. $H_0 = 1$, $H_1 = 2u$, $H_2 = 4u^2 - 2$, Eq. (9.2.33)), and the Gaussian is even, so

$$\varphi_n(-x) = (-1)^n \varphi_n(x). \quad (4)$$

Hence

$$\varphi_n(0) = 0 \iff n \text{ is odd.} \quad (5)$$

Even states ($n = 0, 2, 4, \dots$) have $\varphi_n(0) \neq 0$ and are therefore *not* admissible. The admissible half-oscillator states come from the **odd** full-oscillator states $n = 1, 3, 5, \dots$

Eigenstates and energies of the half oscillator

Label the half-oscillator states by $m = 0, 1, 2, \dots$ via $n = 2m + 1$. Then

$$\psi_m(x) = \begin{cases} \sqrt{2} \varphi_{2m+1}(x), & x > 0, \\ 0, & x \leq 0, \end{cases} \quad m = 0, 1, 2, \dots \quad (6)$$

The factor $\sqrt{2}$ restores normalization: φ_{2m+1} is odd, so $\int_{-\infty}^{\infty} |\varphi_{2m+1}|^2 dx = 1$ splits evenly and $\int_0^{\infty} |\varphi_{2m+1}|^2 dx = \frac{1}{2}$; multiplying by $\sqrt{2}$ gives $\int_0^{\infty} |\psi_m|^2 dx = 1$.

The energies are the odd levels of the full oscillator:

$$E_m = \hbar\omega\left((2m+1) + \frac{1}{2}\right) = \hbar\omega\left(2m + \frac{3}{2}\right), \quad m = 0, 1, 2, \dots \quad (7)$$

i.e. $E = \frac{3}{2}\hbar\omega, \frac{7}{2}\hbar\omega, \frac{11}{2}\hbar\omega, \dots$

Checks

- **Boundary condition satisfied.** Each ψ_m vanishes at $x = 0$ by construction (odd states have a node there); $\psi'_m(0) \neq 0$, which is allowed at an infinite wall.
- **Ground-state energy raised.** The lowest level is $\frac{3}{2}\hbar\omega$, above the full oscillator's $\frac{1}{2}\hbar\omega$. Confining the particle to half the space (forcing a node at the origin) increases the zero-point energy — the expected effect of stronger confinement.
- **Level spacing doubled.** Only every other full-oscillator level survives, so adjacent half-oscillator levels are separated by $2\hbar\omega$ rather than $\hbar\omega$.
- **Same equation, same φ 's.** No new differential equation had to be solved: the half-oscillator spectrum is a *subset* of the full one, and its eigenfunctions are the odd φ_n rescaled by $\sqrt{2}$ on $x > 0$.

Half oscillator = odd states ($n = 1, 3, 5, \dots$) of the full oscillator, $E = \hbar\omega(2m + \frac{3}{2})$.
